

AZ-900: Azure Fundamentals

Domain 3 – Management and Governance + Exam Prep

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Azure Policy vs RBAC

Azure Policy – WHAT

Controls **what can be deployed**. Enforces rules on resources themselves, regardless of who is acting.

- Effect: **Deny** — blocks non-compliant resource creation
- Effect: **Audit** — logs violations without blocking
- Effect: **DeployIfNotExists** — auto-deploys required configs

Example: Policy blocks any resource created without a CostCentre tag.

RBAC – WHO

Controls **who can take action**. Grants or restricts user permissions at subscription, resource group, or resource scope.

- Role assignment = identity + role + scope
- Built-in roles: Owner, Contributor, Reader
- Custom roles are supported

Example: RBAC controls whether a user can create resources at all.

📌 **TESTED:** Policy = WHAT can be deployed. RBAC = WHO can deploy. These are completely separate controls that work together.

Resource Locks

CanNotDelete

Users can read and modify a resource, but **cannot delete it**. The resource must be explicitly unlocked before deletion is possible.

ReadOnly

Users can read the resource but **cannot modify or delete it**. Equivalent to giving everyone the Reader role — no changes of any kind.

Lock Inheritance

Locks applied at a **parent scope cascade downward**: Subscription → Resource Group → Resource. Child resources inherit all parent locks.

- ❏ **CRITICAL:** Locks **override RBAC**. Even a subscription Owner cannot delete a locked resource without first removing the lock. This is heavily tested.

Tags

What Tags Are

Tags are **key-value metadata pairs** applied to Azure resources. They do not affect functionality — they are purely informational labels you attach to resources for organisation and tracking.

- Example: Environment: Production
- Example: CostCentre: Marketing
- Tags are **not inherited** by child resources by default
- Not enforced automatically — tagging is optional unless a Policy enforces it

Why Tags Matter on the Exam

Primary use case: cost allocation. Tags let you group and filter spend by department, project, or environment in Cost Management.

To make tagging mandatory, you must apply an **Azure Policy with Deny effect** — tags alone don't enforce themselves.

Know the difference: tags *label* resources. Policy *enforces* the labelling.

Azure Blueprints

Blueprints package multiple governance components into a **single repeatable deployment unit**. Instead of applying policy, RBAC, and templates separately each time you spin up an environment, Blueprints bundle them together and apply them consistently.

1

Policy Assignments

Enforce compliance rules on every environment from day one — no manual policy wiring required.

2

RBAC Role Assignments

Pre-assign roles to identities so the right people have the right access automatically.

3

ARM Templates

Deploy the actual infrastructure — VNets, storage accounts, VMs — as part of the same Blueprint.

Use case: Standing up pre-configured, governance-compliant environments at scale — governance baked in from day one, not bolted on afterwards.

Cost Tools – Three Stages, Three Tools

Pricing Calculator

Before deployment. Estimate the expected monthly cost of Azure services before you commit. Configure services, regions, and tiers to build a cost estimate from scratch.

TCO Calculator

Comparison tool. Compare your current on-premises infrastructure costs against the equivalent Azure setup. Helps justify cloud migration with a financial case.

Cost Management

After deployment. Analyse and monitor *actual* spend on your live Azure resources. Set budgets, view breakdowns by service or tag, and configure alerts.

📌 **TESTED:** These are three distinct tools. Know which stage each belongs to: Before → Pricing Calculator | Compare → TCO | After → Cost Management.

Cost Reduction Options

Reservations

Commit to **1 or 3 years** of usage on a specific VM size/region. Up to **72% off** pay-as-you-go. Best for stable, always-on workloads — predictable demand only.

Spot VMs

Access unused Azure capacity at up to **90% off**. Workloads can be **evicted with 30 seconds notice** when capacity is reclaimed. Best for batch jobs and fault-tolerant processing.

Savings Plans

Commit to a **fixed hourly spend** across any compute (VMs, containers, functions). More flexible than Reservations — applies across regions, sizes, and services.

Azure Hybrid Benefit

Bring your existing **Windows Server or SQL Server licences** to Azure. Eliminates the software licence cost on top of compute. Not the same as Hybrid Cloud — this is about licences.

Azure Monitor vs Service Health

Azure Monitor – YOUR Resources

Collects metrics and logs from **your own Azure resources**. Lets you set alerts, visualise performance, and query log data.

- Tracks CPU, memory, IOPS, network
- Triggers alerts when thresholds are breached
- Integrates with Application Insights for app telemetry
- Queries via Log Analytics (KQL)

Service Health – AZURE Platform

Reports on the health of **Azure's own infrastructure** — not your resources. Shows outages, planned maintenance, and service advisories that may affect you.

- Active incidents and root cause reports
- Planned maintenance notifications
- Health history for your subscriptions
- Configure alerts for platform events

📌 **TESTED:** Monitor = your resources. Service Health = Azure platform. Completely different. Do not confuse them.

Azure Monitor – Detail



Metrics

Numerical time-series data — CPU%, memory usage, IOPS, network throughput. **Collected automatically** from most Azure resources with no configuration needed.



Log Analytics

Centralised log store. Query log data using **KQL (Kusto Query Language)**. You will not need to write KQL on the AZ-900 exam — just know what Log Analytics is for.



Alerts + Action Groups

Alerts fire when a metric crosses a defined threshold. **Action Groups** define the response — who gets notified and by what method (email, SMS, webhook, ITSM).



Application Insights

APM (Application Performance Monitoring) for web apps. Tracks request rates, response times, error rates, and dependencies. Part of Azure Monitor.

Azure Advisor

Advisor is a **free, personalised recommendation engine** built into Azure. It analyses your resource configuration and usage data and surfaces actionable recommendations — but it does not enforce anything. Recommendations only.



Cost

Identifies idle or undersized resources and suggests rightsizing or shutdown.



Security

Flags missing security configurations and feeds into Defender for Cloud.



Reliability

Recommends HA configurations and redundancy improvements.



Performance

Identifies bottlenecks and suggests scaling or caching improvements.



Operational Excellence

Highlights process and deployment best practice improvements.



TESTED: Advisor RECOMMENDS. Policy ENFORCES. These are not interchangeable.

Microsoft Defender for Cloud

What It Does

Defender for Cloud provides **security posture management** and **threat protection** across your workloads. It was formerly called Azure Security Centre — same product, renamed.

Continuously assesses your environment against security best practices and gives you a consolidated security view.

Key Concepts to Know

- **Secure Score** — a percentage reflecting how many security recommendations you've implemented. Higher = better posture.
- **Multi-cloud and hybrid** — covers Azure, AWS, GCP, and on-premises servers via Azure Arc.
- **Recommendations** — actionable steps to improve your score.
- **Alerts** — threat detections and suspicious activity notifications.

SLAs – Service Level Agreements

An SLA is Microsoft's **contractual uptime guarantee** for a given Azure service, expressed as a percentage (e.g., 99.9%, 99.99%). Missing the SLA threshold entitles you to service credits — but it doesn't mean your app stays up.

Composite SLA

When your architecture chains multiple services together, **multiply the individual SLAs** to get the composite SLA. More dependencies = lower overall availability guarantee. Always lower than any individual SLA in the chain.

Improving Composite SLA

Add **redundancy** — deploying services across multiple availability zones or regions improves your effective uptime above any single-instance SLA.

Free Tiers – No SLA

Free tier services carry zero SLA. Microsoft makes no uptime commitment on free tiers. This is a frequent exam trap — don't assume a free service has any guarantee.

ARM and Infrastructure as Code

Azure Resource Manager (ARM)

ARM is the **management layer** that sits behind every Azure interaction. Whether you use the portal, CLI, PowerShell, or REST API — every request passes through ARM. It handles authentication, authorisation, and resource lifecycle.

Think of ARM as the control plane for all of Azure.

Infrastructure as Code

ARM Templates (JSON) and **Bicep** are IaC tools for declaratively defining Azure infrastructure. Instead of clicking through the portal, you describe what you want and ARM makes it happen.

- Declarative — describe the end state, not the steps
- Repeatable — deploy the same config every time
- Version-controlled — store in Git like any code

Exam note: Know what they are and why they're used. You will not need to read or write JSON or Bicep syntax.



ARCHITECTURE

Azure Arc

Azure Arc **extends Azure's management plane** beyond Azure itself. It lets you manage on-premises and multi-cloud resources as if they were native Azure resources — from the Azure portal, using Azure tools.

1

Extend Azure Policy

Apply the same Policy rules and compliance standards to on-premises servers and non-Azure environments.

2

Extend RBAC + Monitor

Assign roles and collect metrics and logs from Arc-enabled servers, Kubernetes clusters, and databases outside Azure.

3

Unified Management

Single portal view across Azure, AWS, GCP, and on-prem. Manage non-Azure infrastructure without leaving the Azure console.

Well-Architected Framework – 5 Pillars

The Azure Well-Architected Framework is Microsoft's set of guiding principles for building high-quality cloud workloads. Know all five pillars and what each one focuses on.

Reliability

Design for failure. Assume components will fail and architect so the system recovers automatically. Use redundancy, retries, and availability zones.

Security

Protect data, identities, and infrastructure against threats. Apply defence-in-depth, least privilege, and encryption at every layer.

Cost Optimisation

Eliminate waste. Right-size resources, use appropriate pricing models, and monitor spend continuously to avoid overprovisioning.

Operational Excellence

Run and monitor effectively. Automate deployments, implement monitoring, and continuously improve operational processes.

Performance Efficiency

Scale to meet demand. Match resource capacity to actual load — scale out under pressure, scale back when demand drops.

Domain 3 Exam Traps

These are the pairs most commonly confused on the AZ-900. If you get one of these wrong, it's almost always because you picked the right category but the wrong tool. Memorise these distinctions.

Policy vs RBAC

Policy = **WHAT** can be deployed (resource rules). **RBAC** = **WHO** can deploy (user permissions). Completely separate controls.

Pricing Calculator vs Cost Management

Pricing Calculator = **Before** deployment (estimate). **Cost Management** = **After** deployment (actual spend).

Azure Monitor vs Service Health

Monitor = **your resources**. **Service Health** = **Azure platform**. One is about your stuff. One is about Microsoft's infrastructure.

Advisor vs Policy

Advisor recommends. **Policy enforces**. Advisor suggests downsizing. Policy blocks non-compliant deployments.

Resource Locks override RBAC

Even an **Owner cannot delete a locked resource** without removing the lock first. Locks win over role assignments.

Free Tiers + Hybrid Benefit

Free tiers have NO SLA. **Azure Hybrid Benefit = licence reuse** — not Hybrid Cloud (which is a deployment model).

Scenario Practice – Governance

For each scenario, identify the correct service and configuration. The exam presents these as "which service would you use to..." questions.

1 "Prevent resources being deployed outside UK regions"

→ **Azure Policy, Deny effect, Allowed Locations**. Policy enforces where resources can be created.

2 "Remove a leaver's access to Azure resources"

→ **RBAC — remove the role assignment** from the user or their group. Policy has no role here.

3 "Prevent accidental deletion of a production VM"

→ **Resource Lock — CanNotDelete**. Overrides RBAC. Even Owners can't delete without removing the lock.

4 "Find idle VMs and reduce spend"

→ **Azure Advisor, Cost category**. Advisor identifies underused resources and suggests action.

5 "Enforce tagging on all resources in a subscription"

→ **Azure Policy, Deny effect**. Tags don't enforce themselves — Policy must mandate them.

Scenario Practice – Cost

Cost scenario questions typically ask you to pick the right pricing model or tool. Match the scenario clue to the correct answer.

1 "Reduce cost on 100 VMs running 24/7 for the next 3 years"

→ **Reservations**. Predictable, always-on workloads = commit upfront for up to 72% savings.

2 "Cheapest option for a batch processing job that can be interrupted"

→ **Spot VMs**. Interruptible workloads = up to 90% off pay-as-you-go pricing.

3 "Estimate monthly cost before deploying a new environment"

→ **Pricing Calculator**. Always the tool for pre-deployment cost estimation.

4 "Demonstrate cost savings of migrating from on-prem to Azure"

→ **TCO Calculator**. Compares total cost of ownership across on-prem vs cloud.

5 "Identify which department is spending the most on Azure"

→ **Tags + Cost Management**. Tags label resources by department; Cost Management filters spend by tag.

Exam Strategy

In the Exam Room

- Read every question **twice** before answering
- Eliminate the two obviously wrong answers first — narrows it to a coin flip at worst
- Watch for qualifier words: "**MOST appropriate**", "**BEST solution**", "**MINIMUM cost**" — these change the correct answer
- The **simplest service** that meets the stated requirement is almost always correct

Pattern Recognition

"**How to PREVENT X**" → Azure Policy (Deny effect) or Resource Lock.
Prevention = enforcement, not recommendations.

"**Who is responsible for X**" → Shared Responsibility Model. Customer responsibility increases as you move from SaaS → PaaS → IaaS. Microsoft always owns physical infrastructure.

"**Reduce cost on stable workload**" → Reservations. "**Interruptible workload**" → Spot VMs.

What NOT to Memorise

Do NOT Memorise

- Specific pricing figures or thresholds
- CLI commands or PowerShell cmdlet syntax
- ARM template JSON or Bicep syntax
- CIDR notation or subnet calculations
- Specific service quotas or limits
- Low-level networking configuration details

DO Focus On

- **What each service does** and why you'd use it
- **When to choose one over another** — the decision logic
- The relationship between governance tools: Policy, RBAC, Locks, Blueprints
- Cost implications of each pricing model
- **The exam trap pairs** on the previous slide — those are your highest-value revision targets

❏ The AZ-900 tests **conceptual understanding**, not technical configuration. If you can explain what a service does and when to use it, you're ready.